MASTER LINUX. SOLVE PROBLEMS. RUN SYSTEMS **LIKE A PRO.**

ADVANCED LINUX ADMINISTRATION & TROUBLESHOOTING GUIDE

PART 1

By Ramesh Singh

```
ramesh@linux-server:~$ ls -l
total 24
drwxr-xr-x  4 ramesh ramesh  4096 May 20
-rw-r--r--  1 ramesh ramesh   220 May 20
-rw-r--r--  1 ramesh ramesh  3580 May 20
drwxr-xr-x  2 ramesh ramesh  4096 May 19
ramesh@linux-server:~$
```

A PRACTICAL GUIDE FOR SYSADMINS, DEVOPS, AND LINUX ENGINEERS

WHAT YOU WILL LEARN

SYSTEM MANAGEMENT
User, group, process management, permissions, and system monitoring.

STORAGE & FILESYSTEMS
Disk management, LVM, RAID, mounting, fstab, and troubleshooting.

PERFORMANCE TUNING
CPU, memory, I/O monitoring and performance optimization.

SECURITY & ACCESS CONTROL
Permissions, ACL, SELinux, AppArmor, firewalls, and hardening.

NETWORKING
TCP/IP, DNS, firewalls, ports, network troubleshooting.

LOGS & TROUBLESHOOTING
Log analysis, systemd services, debugging and real-world solutions.

REAL-WORLD EXAMPLES · **PRACTICAL COMMANDS** · **EASY TO UNDERSTAND** · **BOOST YOUR CAREER**

"THE BEST WAY TO PREDICT THE FUTURE IS TO CREATE IT"
— WITH LINUX

PART 1
THE FOUNDATION

https://www.linkedin.com/in/rameshsinghit

Advanced Linux Administration & Troubleshooting Guide

1. What happens internally when you run a command in Linux?

Shell reads command → finds path → creates process → kernel executes.

```
which ls
ps -ef | grep command
```

2. Explain the Linux boot process step-by-step.

BIOS → GRUB → Kernel → systemd → services → login

```
dmesg | less
systemctl list-units --type=service
```

3. Difference between process and thread with real use cases.

Process = full program
Thread = part of program

```
ps -eLf
top -H
```

4. What is the role of /proc and /sys file systems?

/proc = system/process info
/sys = hardware info

```
cat /proc/cpuinfo
ls /sys/class
```

5. How does Linux handle memory management?

Uses RAM + swap + cache

```
free -m
vmstat
```

6. What is OOM Killer and when does it trigger?

Kills process when memory is full

```
dmesg | grep -i kill
```

7. Explain context switching and its impact on performance.

CPU switches tasks → too many = slow

```
vmstat 1
```

8. A server is slow — how do you troubleshoot step-by-step?

Check CPU → memory → disk → network

```
top
free -m
iostat
```

9. High CPU usage — how do you identify the root cause?

Find high CPU process

```
top
ps aux --sort=-%cpu
```

10. A process is stuck in D (uninterruptible sleep) state — what will you do?

Check disk/NFS issue

```
ps -eo state,pid,cmd | grep D
iostat
```

11. How do you monitor real-time system performance?

```
top
htop
vmstat 1
```

12. Difference between top, htop, vmstat, iostat, sar?

Use based on need

```
sar -u 1 5
iostat -x 1
```

13. How do you check memory leaks in a running process?

```
top
pmap -x <pid>
```

14. Load average is high but CPU is idle — why?

I/O wait issue

```
iostat -x
```

15. Explain inode and its importance.

Stores file info

```
df -i
```

16. Disk is full but no large files visible — how will you debug?

Deleted files still open

```
lsof | grep deleted
```

17. Difference between soft link vs hard link (practical use case).

Soft = shortcut
Hard = same inode

```
ln file1 file2
ln -s file1 link1
```

18. What is LVM? How do you extend a running filesystem?

Flexible storage

```
lvextend -L +5G /dev/vg/lv
resize2fs /dev/vg/lv
```

19. Explain mount points and /etc/fstab.

Mount = attach disk

```
mount
cat /etc/fstab
```

20. What is journaling in file systems?

Prevents data loss

```
tune2fs -l /dev/sda1
```

21. How do you recover a corrupted filesystem?

```
fsck /dev/sda1
```

22. How does DNS resolution work in Linux?

Local → DNS server

```
cat /etc/resolv.conf
nslookup google.com
```

23. Difference between netstat, ss, and lsof?

```
ss -tulnp
lsof -i :80
```

24. How do you debug a network connectivity issue?

```
ping google.com
traceroute google.com
```

25. A port is not accessible — how will you troubleshoot?

```
ss -tulnp | grep 80
firewall-cmd --list-all
```

@Ramesh Singh

https://www.linkedin.com/in/rameshsinghit

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